THE SHEAFYNESS CONDITION IS EQUIVALENT TO THE RIGHT ADJOINT CARRYING SUPPORTED OBJECTS TO SUPPORTED OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0A9N (sources/stacks-duality/pinned-original-source.tex line 1035).

Allowed standard prerequisites (editorial).

- The right adjoint of pushforward and the sheafyness condition. Same chapter, equation labelled equation-sheafy, invoked in the statement, with the right adjoint a of Rf_* as in the statement. Fixes the condition being characterised.
- Quasi-coherence of the direct image and the notion of being supported on a closed subset. Derived Categories of Schemes, Lemma labelled perfect-lemma-quasi-coherence-direct-image and Definition labelled perfect-definition-supported-on, cited at proof lines 1058-1062. Make the third term of the triangle quasi-coherent and supported.
- Generation of the complexes supported on a closed subset by a perfect object. Derived Categories of Schemes, Lemma labelled perfect-lemma-generator-with-support, cited at proof lines 1088-1092. Provides E and the generating property used in (3).
- Every object as a derived colimit of a system with cones direct sums of shifts. Derived Categories, Lemma labelled derived-lemma-write-as-colimit, cited at proof lines 1092-1096. Feeds the induction that the subcategory is everything.

Source reference. Original equation 0A9L: sources/stacks-duality/pinned-original-source.tex, source lines 988–988. Same-chapter pointer into the author environment equation at source line 988; invoked at proof line 1072 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 08D5: sources/stacks-chow/tags.txt, source lines 10625–10625. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1062 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 08DA: sources/stacks-chow/tags.txt, source lines 10630–10630. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1065 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9A: sources/stacks-chow/tags.txt, source lines 12943–12943. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1094 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 09SN: sources/stacks-chow/tags.txt, source lines 12361–12361. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1096 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of quasi-compact and quasi-separated schemes. Let a be the right adjoint to $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_Y)$. Let $V \subset Y$ be quasi-compact open with inverse image $U \subset X$.*

1. *For every $Q \in D^+_{QCoh}(\mathcal{O}_Y)$ supported on $Y \setminus V$ the image $a(Q)$ is supported on $X \setminus U$ if and only if (0A9L) is an isomorphism on all K in $D^+_{QCoh}(\mathcal{O}_Y)$.*
2. *For every $Q \in D_{QCoh}(\mathcal{O}_Y)$ supported on $Y \setminus V$ the image $a(Q)$ is supported on $X \setminus U$ if and only if (0A9L) is an isomorphism on all K in $D_{QCoh}(\mathcal{O}_Y)$.*
3. *If a commutes with direct sums, then the equivalent conditions of (1) imply the equivalent conditions of (2).*

Proof. Proof of (1). Let $K \in D^+_{QCoh}(\mathcal{O}_Y)$. Choose a distinguished triangle

$$K \to Rj_*K|_V \to Q \to K[1]$$

Observe that Q is in $D^+_{QCoh}(\mathcal{O}_Y)$ (Derived Categories of Schemes, Lemma 08D5) and is supported on $Y \setminus V$ (Derived Categories of Schemes, Definition 08DA). Applying a we obtain a distinguished triangle

$$a(K) \to a(Rj_*K|_V) \to a(Q) \to a(K)[1]$$

on X. If $a(Q)$ is supported on $X \setminus U$, then restricting to U the map $a(K)|_U \to a(Rj_*K|_V)|_U$ is an isomorphism, i.e., (0A9L) is an isomorphism on K. The converse is immediate.

The proof of (2) is exactly the same as the proof of (1).

Proof of (3). Assume the equivalent conditions of (1) hold. Set $T = Y \setminus V$. We will use the notation $D_{QCoh,T}(\mathcal{O}_Y)$ and $D_{QCoh,f^{-1}(T)}(\mathcal{O}_X)$ to denote complexes whose cohomology sheaves are supported on T and $f^{-1}(T)$. Since a commutes with direct sums, the strictly full, saturated, triangulated subcategory $\mathcal{D}$ with objects

$$\{Q \in D_{QCoh,T}(\mathcal{O}_Y) \mid a(Q) \in D_{QCoh,f^{-1}(T)}(\mathcal{O}_X)\}$$

is preserved by direct sums and hence derived colimits. On the other hand, the category $D_{QCoh,T}(\mathcal{O}_Y)$ is generated by a perfect object E (see Derived Categories of Schemes, Lemma 0A9A). By assumption we see that $E \in \mathcal{D}$. By Derived

Categories, Lemma 09SN every object Q of $D_{QCoh,T}(\mathcal{O}_Y)$ is a derived colimit of a system $Q_1 \to Q_2 \to Q_3 \to \ldots$ such that the cones of the transition maps are direct sums of shifts of E. Arguing by induction we see that $Q_n \in \mathcal{D}$ for all n and finally that Q is in $\mathcal{D}$. Thus the equivalent conditions of (2) hold. $\square$